

## The Local Winds

\*Local winds are generated as are the result of pressure and temperature gradient in a particular region and hence affect a very small area. These winds are confined to the lower level of the troposphere.

**Important local winds are as Follows :**

\***Chinook** means ice-eater. These winds descend along the eastern slope of the Rocky Mountains and are warm and dry in nature. These winds increase the temperature of the region during winters resulting in the melting of ice and the emergence of fresh greenery in the region. The winds rising along the southern slope of Alps descend along the northern slope and are warm and dry in nature. It is called **Foehn** in Europe and has its maximum influence in Switzerland.

\* **Santa Ana** is a local wind prevailing in Southern California. It is also a warm and dry wind like Chinook but unlike Chinook, it blows as a Channel Wind in the valley east to west. \***Harmattan** is a local wind that blows in a northeast and east direction in the western parts of Sahara Desert. It is also dry and dusty air. With the advent of Harmattan, the weather becomes dry, pleasant and healthy. It is the reason why Harmattan is also called **Doctor Wind** in the coastal areas of Guinea.

\***Sirocco** is a local wind that blows from the Sahara Desert towards the Mediterranean Sea. The wind is dry, warm and sandy in nature. This air contains a lot of red sand. When blowing over the Mediterranean Sea the wind absorbs moisture and occasionally provides the southern Italy region with rainfall. When red sand descends along with rain it is known as **Blood Rain**. The sand carried by Sirocco from Libya is transported to Malta, Italy, Spain, Croatia, Montenegro, Albania and Greece.

\***Simoom** is a local wind that blows in the deserts of Arabia and Kurdistan. \***Mistral** is a cold polar wind that blows around the northwestern part of the Roun Sea, affecting the areas of Spain and France.

\***Bora** is a dry cold air that moves at a very fast pace. It prevails along the eastern coasts of Adriatic Sea. The northern part of Italy is highly influenced by Bora.

\***Loo** is a dry and warm local wind that prevails in northern India. It blows from northwest and west to the east. These winds are tropical and dry and travel at a high velocity.

\***Norwesters** are pre monsoon thunderstorms blowing in India which are also known as **Kalbaisakhi**. \***Zonda** is a warm dry wind that blows along the eastern slopes of the

Andes Mountains and influences the country of Argentina.

\***Black Roller** is a local wind prevailing in the Great Plains region of North America. \***Purga** is a cold local wind prevailing in the tundras of Russia. \***Levanter** is a cold local wind that blows from east to west in Spain.

\***Pampero** is a local northwestern cold wind of the Pampas region of South America.

1. Match List-I with List-II and select the correct answer using the codes given below the lists.

**List-I (Local Wind)**

- |              |           |
|--------------|-----------|
| A. Foehn     | B. Simoom |
| C. Santa Ana | D. Zonda  |

**List-II (Region)**

- |               |              |
|---------------|--------------|
| 1. Argentina  | 2. Kurdistan |
| 3. California | 4. Alps      |

**Code :**

|     | A | B | C | D |
|-----|---|---|---|---|
| (a) | 2 | 4 | 1 | 3 |
| (b) | 4 | 2 | 3 | 1 |
| (c) | 2 | 4 | 3 | 1 |
| (d) | 4 | 2 | 1 | 3 |

**I.A.S. (Pre) 2001**

**Ans. (b)**

Foehn- A warm dry wind blowing down the northern slopes of the Alps. It most influences the weather of Switzerland.  
 Simoom- A strong, dry wind that blows in the deserts of Arabia and Kurdistan.  
 Santa Ana - The local wind of southern California.  
 Zonda - The local wind that blows on the eastern slope of the Andes in Argentina.

2. 'Foehn' is a local wind of –

- |           |                 |
|-----------|-----------------|
| (a) China | (b) Korea       |
| (c) Japan | (d) Switzerland |

**U.P.U.D.A./L.D.A. (Pre) 2010**

**Ans. (d)**

See the explanation of the above question.

3. Which one of the following pairs is not correctly matched?

- |             |                 |
|-------------|-----------------|
| (a) Foehn   | - Alps Mountain |
| (b) Bora    | - Poland        |
| (c) Mistral | - Rhine Valley  |
| (d) Khamsin | - Egypt         |

**U.P.P.C.S. (Pre) 2016**

**Ans. (b)**

Bora is a dry, cold and fast blowing wind on the eastern coast of the Adriatic Sea. The northern part of Italy is particularly influenced by this wind. Thus, option (b) is not correctly matched. Other options are correctly matched.

4. Which one of the following is not correctly matched?

- (a) Chinook-U.S.A. (b) Sirocco-Sicily  
(c) Blizzard-Chile (d) Norwesters-India

U.P.P.C.S. (Pre) 2005

Ans. (c)

**Chinook** - It is a local wind that blows in the east of the Rockies mountains, USA and Canada.

**Sirocco** - A wind that blows from the Sahara Desert toward the northeast direction of the Mediterranean Sea and enters Sicily (Italy), Spain and other European regions/countries.

**Blizzard** - A blizzard is a long-lasting snowstorm with very strong winds and intense snowfall. It flows through South Pole, Siberia, Canada, and the U.S.A. Chile does not fall on its way.

**Norwester** - It is a pre-monsoon thunderstorm of India also called 'Kalbaisakhi'.

5. Which one of the following locations has "Blood Rain".

- (a) Italy (b) France  
(c) Canada (d) Spain

U.P. Lower Sub. (Pre) 1998

Ans. (a)

Sirocco is a warm, dry and dusty wind blowing from Sahara Desert to the Mediterranean Sea. This wind carries red sand with it. This wind sometimes causes rain in Italy. When the red sand falls to the earth with the raindrops, it is called Blood Rain.

6. Consider the following pairs :

(Name of the local Winds) (Related country)

1. Sirocco - France  
2. Bora - Italy  
3. Blizzard - Canada

Which one of the above pairs is not correctly matched ?

- (a) 1 only (b) 2 only  
(c) 2 and 3 only (d) 3 only

U.P.P.C.S. (Mains) 2013

Ans. (a)

Sirocco is a Mediterranean wind that blows from the Sahara Desert. Sirocco carries dust from Libya to Malta, Italy, Croatia, Montenegro, Albania and Greece. Therefore, it is not related to France.

7. What about the Chinook winds over the Central plains of the United state what effect?

- (a) The temperature of winter increases.  
(b) The temperature of summer decreases.  
(c) Remain the same temperature.  
(d) There is no effect on temperature.

Jharkhand P.C.S. (Pre) 2013

Ans. (a)

The meaning of 'Chinook' is 'Snow Eater'. It is a warm and dry wind descending along the eastern slopes of the Rocky Mountains, blowing from the southern part of Colorado to the Southern part of British Columbia in Canada. Due to its effects, the temperature in summer increases snow melts and green grass grows even in winter. It is beneficial for cattle rearers, as it makes the pasture snow-free.

8. Which of the following is not correctly matched ?

- | (Wind)        | (Country)  |
|---------------|------------|
| (a) Santa Ana | California |
| (b) Haboob    | Sudan      |
| (c) Yamo      | Japan      |
| (d) Mistral   | Australia  |

U.P.P.C.S. (Pre) 2019

Ans. (d)

The correct match of country and wind is :

- | Wind      | Country/State    |
|-----------|------------------|
| Santa Ana | California       |
| Haboob    | Sudan            |
| Yamo      | Japan            |
| Mistral   | France and Spain |

So, Mistral is not correctly matched. The correct answer is option (d).

9. Which one of the following pairs is NOT correctly matched?

- | (Name of local winds) | (Places)      |
|-----------------------|---------------|
| (a) Leveche           | Spain         |
| (b) Brickfielder      | Australia     |
| (c) Black roller      | North America |
| (d) Shamal            | Austria       |

U.P.P.C.S. (Pre) 2021

Ans. (d)

Shamal is a hot, dry and dusty wind blowing mainly from June to August, which blows to the north or northwest in Iraq, Iran and the Arabian Peninsula. Its speed is approx. 45 miles per hour.